Notes on Categories

Apiros3

First Version : September 16, 2025
Last Update : – –, 2025

Contents

1 Introduction **2**

2 Basics **3**
2.1 Object Definitions . . . 3
2.1.1 Functor . . . 3
2.1.2 Natural Transformation . . . 4
2.1.3 Monos, Epis, Zeros . . . 5
2.1.4 Equivalence . . . 5
2.1.5 Examples to keep in mind . . . 6
2.2 Constructing Categories . . . 6
2.2.1 Contravariance . . . 6
2.2.2 Product . . . 6

3 Universals, Limits, and Adjoints **7**
3.1 Universals . . . 7
3.2 Limits and Colimits . . . 8
3.2.1 Coproducts . . . 8
3.2.2 Coequaliser . . . 9

1 Introduction

This note aims to put an uncondensed summary of concepts related to Category Theory I've come across. Hopefully full of examples, as this should have what I needed to gain an intuition of the concepts. The notes here are (so far) based off of:

1. C2.7: Category Theory by Pavel Safronov (Oxford Maths Cateogry Theory Course Notes)
2. Introduction to Categories and Categorical Logic by Samson Abramsky and Nikos Tzevelekos (Oxford CS Category Proofs and Processes Course Notes)
3. Categories for the Working Mathematician (Second Edition, only sections 1 to 6) by Saunders Mac Lane
4. Category Theory by Steve Awodey

2 Basics

2.1 Object Definitions

Definition 2.1.1. *A category $\mathcal{C}$ consists of the following data:*

- *A collection* $\text{ob}\ \mathcal{C}$ *of objects of* $\mathcal{C}$
- *For every* $x, y \in \text{ob}\ \mathcal{C}$ *a collection* $\text{Hom}_{\mathcal{C}}(x, y)$ *of morphisms*
- *For every* $x \in \text{ob}\ \mathcal{C}$*, the identity morphism* $\text{id}_x \in \text{Hom}_{\mathcal{C}}(x, x)$
- *For every* $x, y, z \in \text{ob}\ \mathcal{C}$*, the composition map*

$$\circ : \text{Hom}_{\mathcal{C}}(y, z) \times \text{Hom}_{\mathcal{C}}(x, y) \to \text{Hom}_{\mathcal{C}}(x, z)$$

These then must satisfy the following axioms:

- *For any two* $x, y \in \text{ob}\ \mathcal{C}$ *and any morphism* $f \in \text{Hom}_{\mathcal{C}}(x, y)$ *we have*

$$f \circ \text{id}_x = f \qquad \text{id}_y \circ f = f$$

- *Morphisms under composition are associative*

Notation 2.1.2. We write $x \in \mathcal{C}$ for $x \in ob\ \mathcal{C}$ and omit the subscript in *Hom* when the category is clear. We may sometimes write $\mathcal{C}(x, y)$ for $Hom_{\mathcal{C}}(x, y)$. We also write $Hom_{\mathcal{C}}(x)$ for $Hom_{\mathcal{C}}(x, x)$ and call these endomorphisms.

2.1.1 Functor

Definition 2.1.3. *A functor* $F : \mathcal{C} \to \mathcal{D}$ *between two categories* $\mathcal{C}, \mathcal{D}$ *consists of the following data:*

- *a map* $F : \text{ob}\ \mathcal{C} \to \text{ob}\ \mathcal{D}$
- *For any two objects* $x, y \in \mathcal{C}$*, a map of sets* $F : \text{Hom}_{\mathcal{C}}(x, y) \to \text{Hom}_{\mathcal{D}}(F(x), F(y))$

Such that they satisfy

- *Unit: for any* $x \in \mathcal{C}$*,* $F(\text{id}_x) = \text{id}_{F(x)}$
- *For any objects* $x, y, z \in \mathcal{C}$ *and morphisms* $f \in \text{Hom}_{\mathcal{C}}(x, y)$ *and* $g \in \text{Hom}_{\mathcal{C}}(y, z)$*, we have*

$$F(g \circ f) = F(g) \circ F(f)$$

We say that a functor $F : \mathcal{C} \to \mathcal{D}$ *is* ***faithful*** *if the map* $\text{Hom}_{\mathcal{C}}(x, y) \to \text{Hom}_{\mathcal{D}}(F(x), F(y))$ *is injective for any objects* x *and* y*. We say that* F *is* ***full*** *if this map is surjective, and that it is* ***fully faithful*** *if it is both full and faithful.*

We say that F *is a* ***contravariant functor*** *from* $\mathcal{C}$ *to* $\mathcal{D}$ *if it is a functor* $\mathcal{C}^{\text{op}} \to D$*. A functor* $\mathcal{C}$ *to* $\mathcal{D}$ *is also referred to as a covariant functor.*

Note we use the same F to refer to the map between both objects and morphisms.

Definition 2.1.4. *A morphism* $f \in \text{Hom}(x, y)$ *in a category is an* ***isomorphism*** *if there is a morphism* $f^{-1} \in \text{Hom}(y, x)$ *such that* $f^{-1} \circ f = \text{id}_x$ *and* $f \circ f^{-1} = \text{id}_y$*. We also say that* f *is* ***invertible****. If two objects* $x, y \in \mathcal{C}$ *are isomorphic, we write* $x \cong y$*.*

Remark 2.1.5. As usual, inverses are unique. Suppose that $f \in Hom(x, y)$ and we have $g \circ f = id_x$ and $f \circ h = id_y$. Then,

$$g = g \circ id_y = g \circ (f \circ h) = (g \circ f) \circ h = id_x \circ h$$

Definition 2.1.6. *A category $\mathcal{C}$ is called a* ***groupoid*** *if every morphism is invertible. We say that a groupoid is* ***connected*** *if any two objects are isomorphic.*

Remark 2.1.7. Consider a groupoid with a single object. The data required to specify such a groupoid is the monoid of endomorphisms in which every object has an inverse. This is just a group.

2.1.2 Natural Transformation

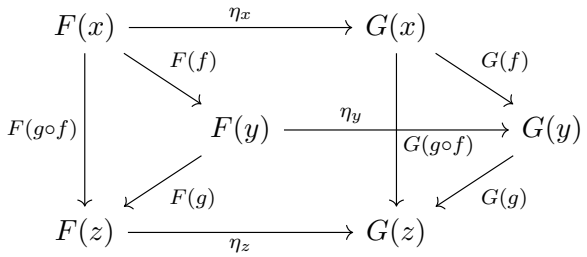
Definition 2.1.8. *Let $F, G : \mathcal{C} \to \mathcal{D}$ be two functors. A natural transformation $\eta : F \Rightarrow G$ consists of morphisms $\eta_x \in \mathrm{Hom}_{\mathcal{D}}(F(x), G(x))$ for every object $x \in \mathcal{C}$ such that the diagram*

$$\begin{array}{ccc} F(x) & \xrightarrow{F(f)} & F(y) \\ {\scriptstyle \eta_x}\downarrow & & \downarrow{\scriptstyle \eta_y} \\ G(x) & \xrightarrow[G(f)]{} & G(y) \end{array}$$

commutes for every morphism $f \in \mathrm{Hom}_{\mathcal{C}}(x, y)$.

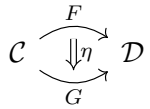
We say a natural transformation $\eta : F \Rightarrow G$ is a ***natural isomorphism*** *if the morphisms η_x are isomorphisms for any $x \in \mathcal{C}$.*

Equivalently, we can also write something like:

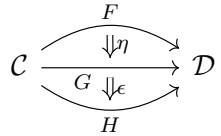


Given two categories $\mathcal{C}$ and $\mathcal{D}$, one can construct a category $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$ of functors between $\mathcal{C}$ and $\mathcal{D}$. It's objects are functors $\mathcal{C} \to \mathcal{D}$ and morphisms are given by natural transformations. Then we can view natural isomorphisms as isomorphisms in the functor category.

Notation 2.1.9. We write natural transformations as



1. Consider



   This gives *vertical composition* $F \Rightarrow H$ with components given by the composition $F(x) \overset{\eta_x}{\to} G(x) \overset{\epsilon_x}{\to} H(x)$ for all $x \in \mathcal{C}$.

2. Consider

$$\mathcal{C} \underset{G_1}{\overset{F_1}{\Downarrow \eta}} \mathcal{D} \underset{G_2}{\overset{F_2}{\Downarrow \epsilon}} \mathcal{E}$$

This gives *horizontal composition* $F_2F_1 \Rightarrow G_2G_1$ given by composition

$$F_2F_1(x) \overset{F_2(\eta_x)}{\to} F_2G_1(x) \overset{\epsilon_{(G_1(x))}}{\to} G_2G_1(x)$$

for every $x \in \mathcal{C}$. Picking η or ϵ to be the identity functor, we can compose by functors as well.

Example 2.1.10. Define $\det_K(M)$ be the determinant of the $n \times n$ matrix with entries in the commutative ring K, and let K^* denote the group of units of K. Thus M is non-singular when $\det_K(M)$ is a unit, so this gives a morphism $\mathrm{GL}_n(K) \to K^*$ of groups. Then, for any morphism $f : K \to K'$ this gives a commutative diagram:

$$\begin{array}{ccc} \mathrm{GL}_n(K) & \xrightarrow{\det_K} & K^* \\ {\scriptstyle \mathrm{GL}_n f}\downarrow & & \downarrow{\scriptstyle f^*} \\ \mathrm{GL}_n(K') & \xrightarrow[\det_{K'}]{} & K'^* \end{array}$$

Showing that the transformation $\det\{K : \mathbf{CRng}\} : \mathrm{GL}_n(K) \to K^*$ between the two functors $\mathbf{CRng} \to \mathbf{Grp}$.

Example 2.1.11. For each group G the projection $p_G : G \to G/[G,G]$ defines a transformation from the identity functor on **Grp** to the factor-commutator functor $\mathbf{Grp} \to \mathbf{Ab} \to \mathbf{Grp}$. Then p is natural, as for any group homomorphism $f : G \to H$ we have the evident homomorphism f' for which the following commutes:

$$\begin{array}{ccc} G & \xrightarrow{p_G} & G/[G,G] \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ H & \xrightarrow[p_H]{} & H/[H,H] \end{array}$$

2.1.3 Monos, Epis, Zeros

TODO!!!: S1.5

2.1.4 Equivalence

Definition 2.1.12. *An **equivalence of categories** $\mathcal{C}, \mathcal{D}$ is a pair of functors $F : \mathcal{C} \to \mathcal{D}$, $G : \mathcal{D} \to \mathcal{C}$ along with natural isomorphisms $e : \mathrm{id}_{\mathcal{C}} \Rightarrow GF$ and $\epsilon : FG \Rightarrow \mathrm{id}_{\mathcal{D}}$.*

Definition 2.1.13. *An **adjoing equivalence** of categories $\mathcal{C}, \mathcal{D}$ is an equivalence (F, G, e, ϵ) satisfying the following:*

1. *The composite natural transformation*

$$F \cong F \circ \mathrm{id}_{\mathcal{C}} \overset{\mathrm{id}_F \circ e}{\Rightarrow} FGF \overset{\epsilon \circ \mathrm{id}_F}{\Rightarrow} \mathrm{id}_{\mathcal{D}} \circ F \cong F$$

is the identity natural transformation on F.

2. *The composite natural transformation*

$$G \cong \mathrm{id}_{\mathcal{C}} \circ G \overset{e \circ \mathrm{id}_G}{\Rightarrow} GFG \overset{\mathrm{id}_G \circ \epsilon}{\Rightarrow} G \circ \mathrm{id}_{\mathcal{D}} \cong G$$

is the identity natural transformation on G

2.1.5 Examples to keep in mind

Example 2.1.14. Groups form a category **Grp** with morphisms by homomorphisms of groups. We can restrict groups to be abelian, which forms a category **Ab**.

We consider some functors:

- The forgetful functor from **Ab** to **Grp** is fully faithful.
- The abelianization $G \mapsto G/[G,G]$ gives a functor $\mathbf{Grp} \to \mathbf{Ab}$. It is neither full nor faithful.

Example 2.1.15. If k is a field, k-vector spaces form a category $\mathbf{Vect}_k$ with morphisms given by linear maps.

Example 2.1.16. (Small) categories form a category **Cat** where morphisms are given by functors.

Example 2.1.17. A set X can be regarded as a category $\mathcal{C}$ with $ob\,\mathcal{C} = X$ where $Hom_{\mathcal{C}}(x,y) = \emptyset$ for $x \neq y$ and $End_{\mathcal{C}}(x) = \{id_x\}$. These categories are called a **discrete category**.

Given a set and viewing it as a discrete category, this gives a fully faithful functor $\mathbf{Set} \to \mathbf{Cat}$.

2.2 Constructing Categories

2.2.1 Contravariance

Definition 2.2.1. *Given a category $\mathcal{C}$, we have the **opposite category** $\mathcal{C}^{\mathrm{op}}$, which has the same objects and $\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(x,y) = \mathrm{Hom}_{\mathcal{C}}(y,x)$.*

2.2.2 Product

Definition 2.2.2. *Given two categories $\mathcal{B}$ and $\mathcal{C}$, the **product** of $\mathcal{B}$ and $\mathcal{C}$, written $\mathcal{B} \times \mathcal{C}$ is a category where objects are pairs $\langle b, c\rangle$ of objects $b \in \mathcal{B}$ and $c \in \mathcal{C}$, and an arrow $\langle b, c\rangle \to \langle b', c'\rangle$ is a pair $\langle f, g\rangle$ of arrows $f : b \to b'$ and $g : c \to c'$, with composition of arrows*

$$\langle b, c\rangle \xrightarrow{\langle f,g\rangle} \langle b', c'\rangle \xrightarrow{\langle f',g'\rangle} \langle b'', c''\rangle$$

is defined by composites in $\mathcal{B}$ and $\mathcal{C}$ by

$$\langle f', g'\rangle \circ \langle f, g\rangle = \langle f' \circ f, g' \circ g\rangle$$

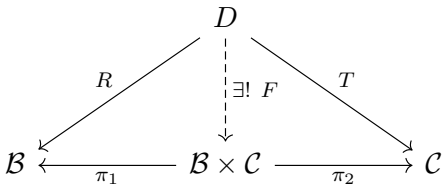
Definition 2.2.3. *Given a product $\mathcal{B} \times \mathcal{C}$, we have functors π_1 and π_2 called projections on the product, defined by*

$$\pi_1\langle f, g\rangle = f, \qquad \pi_2\langle f, g\rangle = g$$

Given any category D and two functors

$$\mathcal{B} \xleftarrow{R} D \xrightarrow{T} \mathcal{C}$$

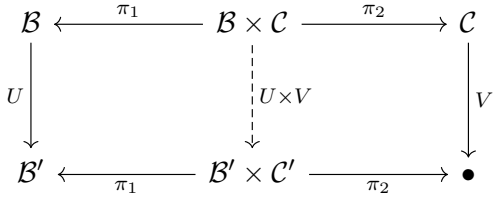
there is a unique functor F such that the following diagram commutes:



Definition 2.2.4. *Given two functors $U : \mathcal{B} \to \mathcal{B}'$ and $V : \mathcal{C} \to \mathcal{C}'$, we have a **product** of functors $U \times V : \mathcal{B} \times \mathcal{C} \to \mathcal{B}' \times \mathcal{C}'$ which is defined explicitly on objects and arrows by*

$$(U \times V)\langle b, c\rangle = \langle U\, b, V\, c\rangle, \qquad (U \times V)\langle f, g\rangle = \langle U\, f, V\, g\rangle$$

Alternatively, this functor is the unique functor that makes the following diagram commutative:



Hence the product $\times$ is a pair of functions that maps pairs of categories to a new category via $\langle \mathcal{B}, \mathcal{C}\rangle \mapsto \mathcal{B} \times \mathcal{C}$ and pairs of functors to a new functor via $\langle U, V\rangle \mapsto U \times V$. When we have composites $U' \circ U$ and $V' \circ V$, we have $(U' \times V') \circ (U \times V) = (U' \circ U) \times (V' \circ V)$. Hence $\times$ itself is a functor with

$$\times : \mathbf{Cat} \times \mathbf{Cat} \to \mathbf{Cat}$$

This overloads to functors like $\mathbf{Grp} \times \mathbf{Grp} \to \mathbf{Grp}$ and $\mathbf{Top} \times \mathbf{Top} \to \mathbf{Top}$, etc.

Definition 2.2.5. *A **bifunctor** is a functor from the product category.*

Example 2.2.6. The cartesian product $X \times Y$ on two sets X and Y is the object function of a bifunctor $\mathbf{Set} \times \mathbf{Set} \to \mathbf{Set}$.

Fixing an argument in the bifunctor gives an ordinary functor. The whole bifunctor is determined by the two arrays of such functors. That is,

Proposition 2.2.7. *Let $\mathcal{B}, \mathcal{C}$ and $\mathcal{D}$ be categories. For all objects $b \in \mathcal{B}$ and $c \in \mathcal{C}$, let*

$$L_c : \mathcal{B} \to \mathcal{D}, \qquad M_b : \mathcal{C} \to \mathcal{D}$$

be functors such that $M_b(c) = L_c(b)$ for all b and c. Then, there exists a bifunctor $S : B \times C \to D$ with $S(-, c) = L_c$ for all c and $S(b, -) = M_b$ for all b if and only if for every pair of arrows $f : b \to b'$ and $g : c \to c'$ one has

$$M_b\, g \circ L_c\, f = L_c\, f \circ M_b\, g$$

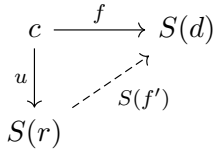
These arrows in $\mathcal{D}$ are then the value of $S(f, g)$ of the arrow function of S at f and g.

Proof. □

3 Universals, Limits, and Adjoints

3.1 Universals

Definition 3.1.1. *If $S : \mathcal{D} \to \mathcal{C}$ is a functor and $c \in \mathcal{C}$, a **universal arrow** from c to S is a pair $\langle r, u\rangle$ consisting of an object $r \in \mathcal{D}$ and an arrow $u : c \to S(r)$ of $\mathcal{C}$ such that for every pair $\langle d, f\rangle$ with an object $d \in \mathcal{D}$ and an arrow $f : c \to F(d)$ of $\mathcal{C}$, there is a unique arrow $f' : r \to d$ of $\mathcal{D}$ with $F(f' \circ u) = f$. Alternatively, every arrow f to S factors uniquely through the universal arrow u:*

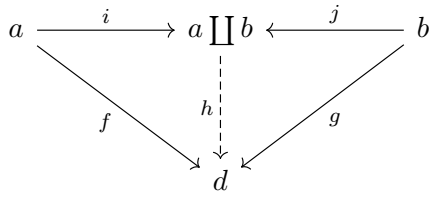


3.2 Limits and Colimits

3.2.1 Coproducts

Definition 3.2.1. *For any category $\mathcal{C}$, define the* ***diagonal functor*** $\Delta : \mathcal{C} \to \mathcal{C} \times \mathcal{C}$ *on objects by* $\Delta(c) = \langle c, c \rangle$ *and on arrows by* $\Delta(f) = \langle f, f \rangle$*. A universal arrow from an object $\langle a, b \rangle$ of $\mathcal{C} \times \mathcal{C}$ to the functor Δ is called a* ***coproduct diagram****.*

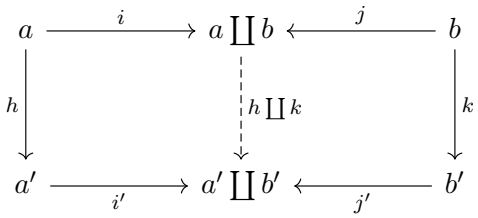
The coproduct diagram consists of an object $a \coprod b$ in $\mathcal{C}$ called the coproduct object, and an arrow $\langle a, b \rangle \to \langle a \coprod b, a \coprod b \rangle$ of $C \times C$, which we can name by a pair of arrows $i : a \to a \coprod b$ and $j : b \to a \coprod b$ called injections. This pair has the universal property such that any maps $f : a \to d$ and $g : b \to d$ factor through $a \coprod b$ uniquely. That is, we can fill the following diagram uniquely



This gives a bijection

$$\hom(a, d) \times \hom(b, c) \cong \hom(a \coprod b, d)$$

by the assignment $\langle f, g \rangle \mapsto h$. This is natural in d, with inverse $h \mapsto \langle h\, i, h\, j \rangle$. If every pair of objects in $\mathcal{C}$ has a coproduct, choosing this gives a bifunctor $\coprod \mathcal{C} \times \mathcal{C} \to \mathcal{C}$, where $h \coprod k$ is defined for arrows $h : a \to a'$ and $k : b \to b'$ as:



The coproduct object is unique up to isomorphism, and exists in many familiar categories:

Set	disjoint union of sets,
Top	disjoint union of spaces,
$\mathbf{Top}_*$	wedge product (join two spaces at the base points),
Ab, R-**Mod**	direct sum $A \oplus B$,
Grp	free product,
CRng	tensor product $R \otimes S$.

In a preorder P, the least upper bound $a \vee b$ of two elements translates exactly to show that it is a coproduct of a and b in P, regarded as a category. The upper bound ensures injections and the leastness ensures the factoring.

Definition 3.2.2. *Fix a set X, consider the functor category $\mathcal{C}^X$ with objects as X-indexed families of $\mathcal{C}$. The diagonal functor $\Delta : \mathcal{C} \to \mathcal{C}^X$ sends each c to the contant family. The universal arrow from a to Δ is an X-fold coproduct diagram; a coproduct object $coprod_x a_x \in \mathcal{C}$ and arrows $i_x : a_x \to \coprod_x a_x$ of $\mathcal{C}$ with the universal property.*

The universal property then states that the assignment $f \mapsto \{f\, i_x \mid x \in X\}$ is a bijection

$$\hom(\coprod_x a_x, c) \cong \prod_x \hom(a_x, c)$$

which is natural in c. For instance in **Set**, a coproduct is an X-fold disjoint union.

Definition 3.2.3. *If the factors in a coproduct are all equal, the coproduct $\coprod_x b$ is called a* ***copower*** *and is written $X \cdot b$.*

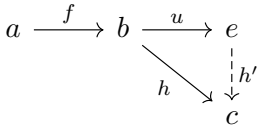
Here, we have

$$\hom(X \cdot b, c) \cong \hom(b, c)^X$$

and is natural in c. For instance in **Set** with $b = Y$ a set, the copower $X \cdot Y = X \times Y$ is the cartesian products of the sets X and Y.

3.2.2 Coequaliser

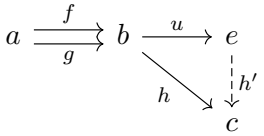
Definition 3.2.4. *Suppose that $\mathcal{C}$ has a null object z such that for any two objects $b, c \in C$ there is a zero arrow $0 : b \to z \to c$. The* ***cokernel*** *of $f : a \to b$ is an arrow $u : b \to e$ such that $uf = 0$ and any arrow $h : b \to c$ such that $hf = 0$ uniquely factors through h' as follows:*



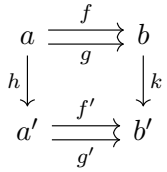
Example 3.2.5. In **Ab**, the cokernel of $f : A \to B$ is the projection $B \to B/f\,A$ to a quotient group of B.

In many categories a cokernel is typically a suitable quotient object, but in categories without a null object cokernels are not available. A more general notion is the following:

Definition 3.2.6. *Let $f, g : a \to b$ over a category $\mathcal{C}$, a* ***coequaliser*** *of $\langle f, g\rangle$ is an arrow $u : b \to e$ such that $uf = ug$, and for any $h : b \to c$ such that $hf = hg$, h factors uniquely through some $h' : e \to c$. The picture is:*



Remark 3.2.7. Let $\downarrow\downarrow$ denote the category with two objects and two non-identity arrows from the first object to the second. Consider the functor category $\mathcal{C}^{\downarrow\downarrow}$. An object in $\mathcal{C}^{\downarrow\downarrow}$ is a functor from $\cdot \rightrightarrows \cdot$ to $\mathcal{C}$; or a pair $\langle f, g\rangle : a \to b$ of parallel arrows $a \rightrightarrows b$ in $\mathcal{C}$. An arrow is just natural transformations between the functors, which is a pair $\langle h, k\rangle$ such that the following f-square and g-square commute.



There is a diagonal functor $\Delta : \mathcal{C} \to \mathcal{C}^{\downarrow\downarrow}$ defined on objects c and arrows r of $\mathcal{C}$ as